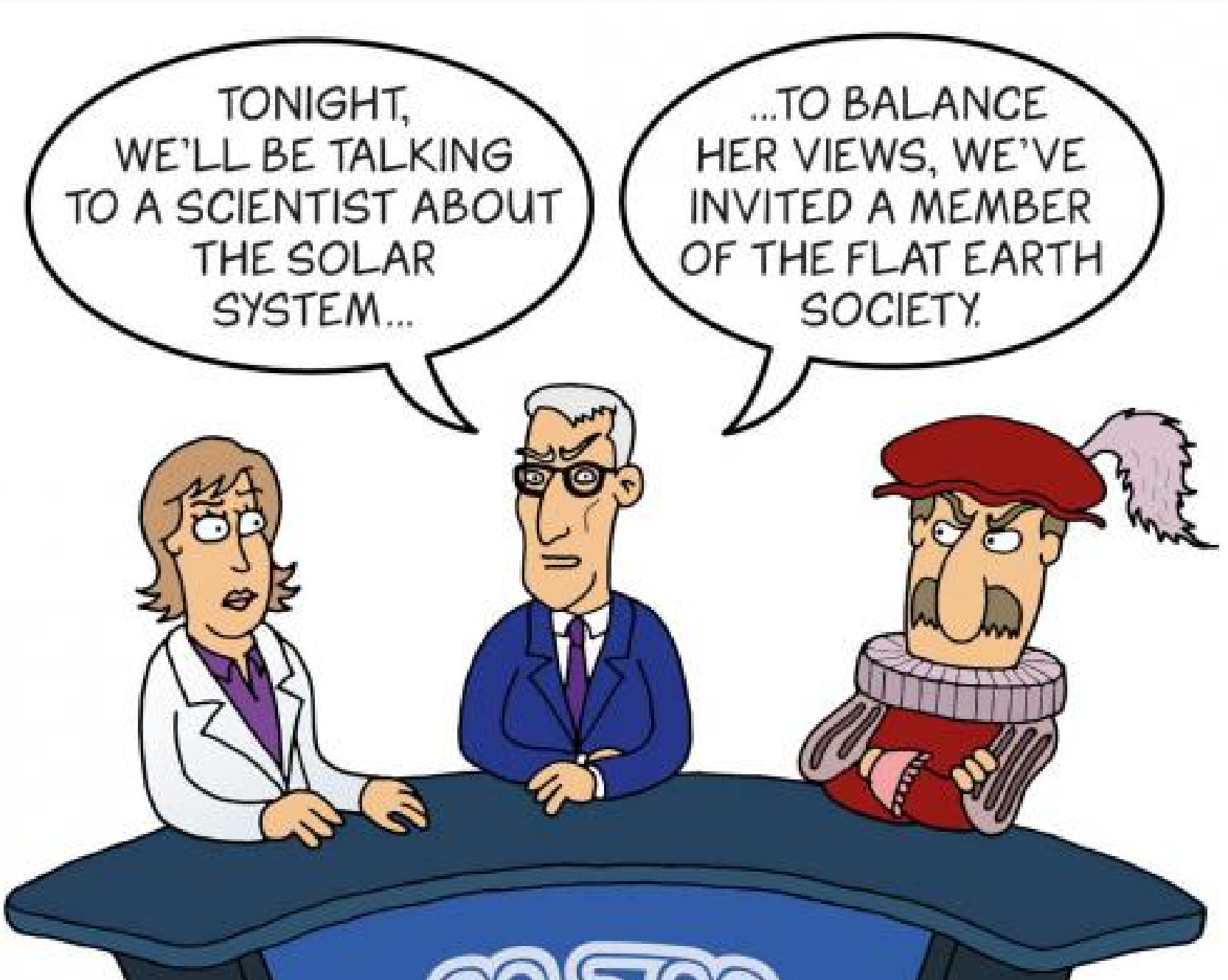
~~Reproducibility Research and Open Science~~ Ethics, Scientific Integrity, Deontology, and ... GenAI

Arnaud Legrand, ADN team (Anthropocene, Degrowth, and ICT)

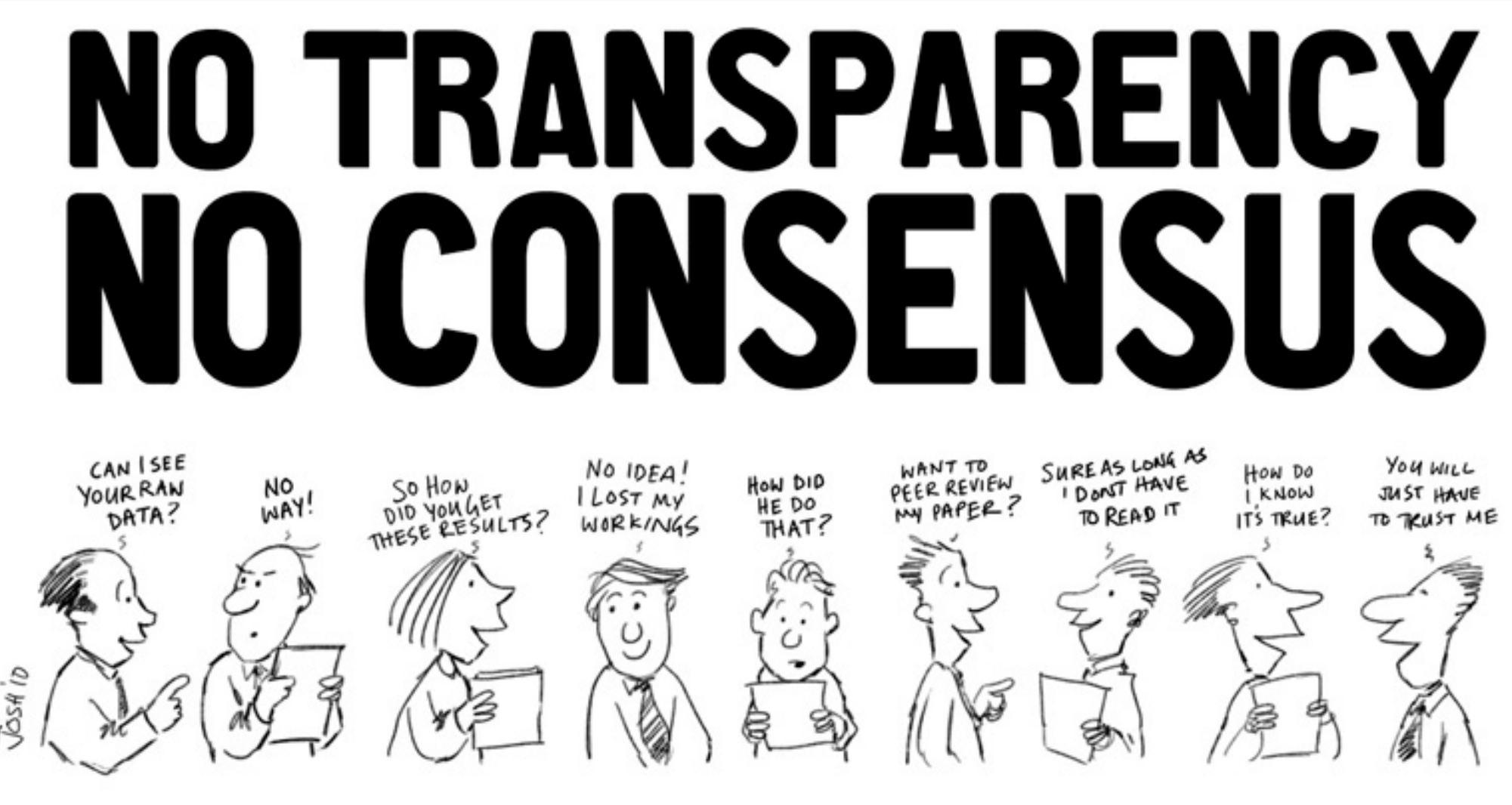
Doing a PhD, good practice and pitfalls to avoid
October 2026



Scientific Consensus



Scientific Consensus



Different Reproducibility Concerns in Modern Science

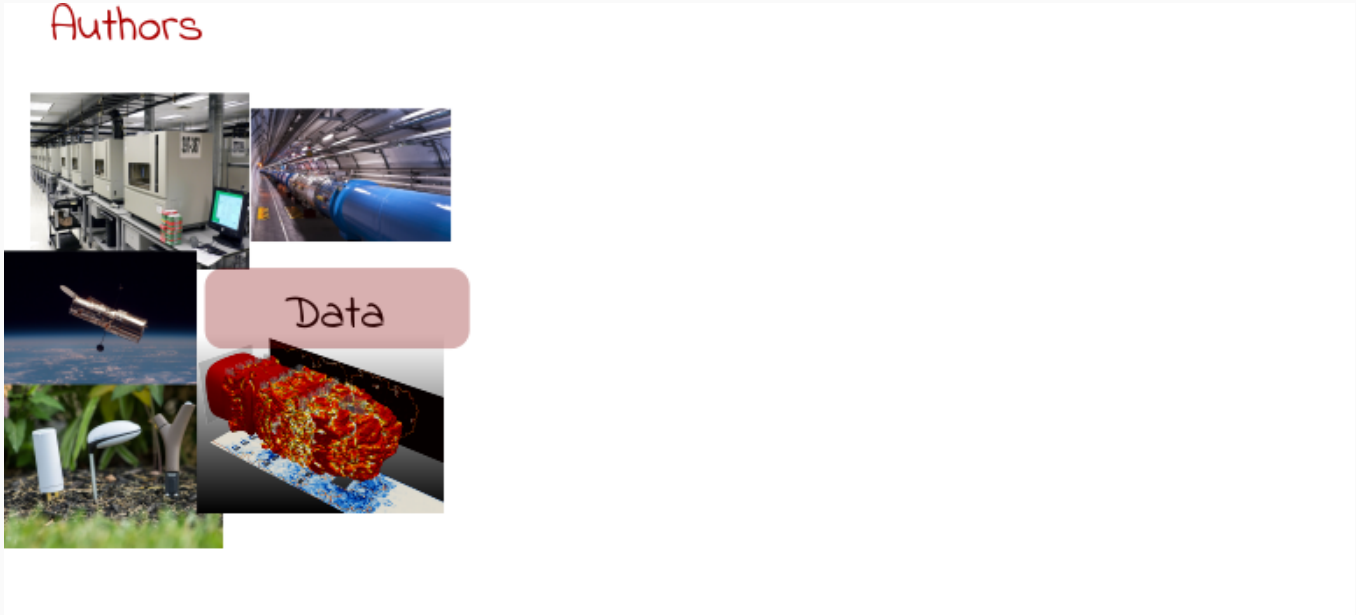
Social Sciences, Oncology, ... methodology, statistics, pre-registration

Genomics software engineering, computational reproducibility, provenance

Computational fluid dynamics numerical issues

Artificial Intelligence most of the above

The processing steps between raw observations and findings have gotten increasingly numerous and complex



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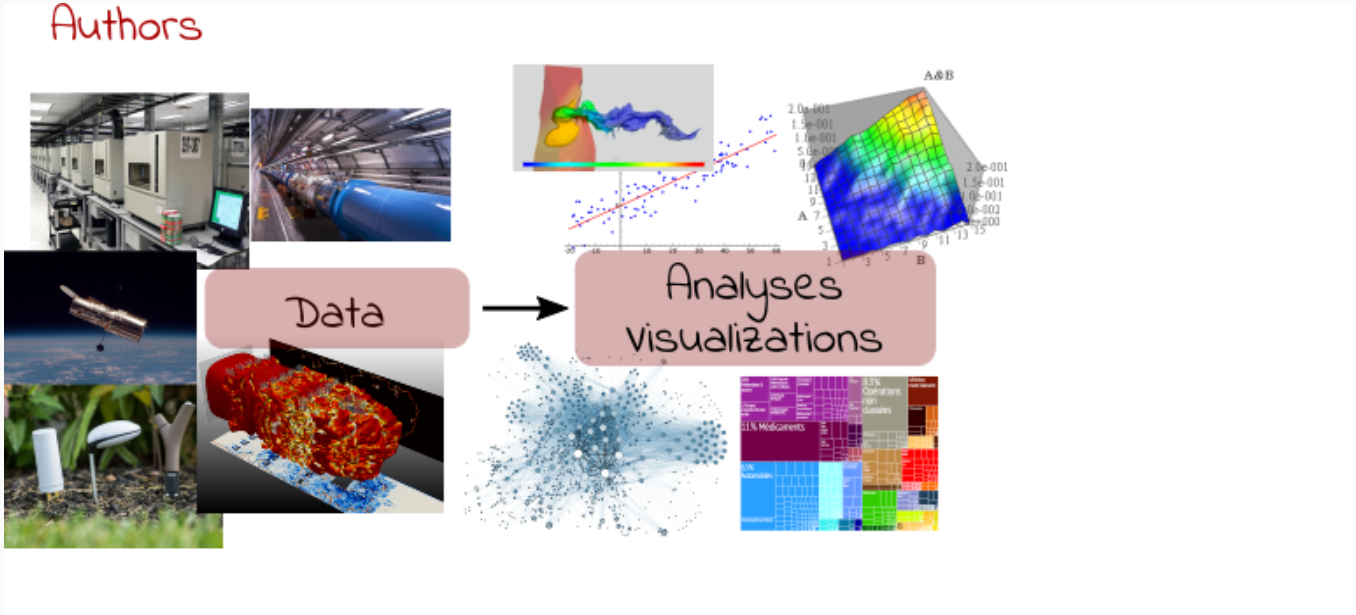
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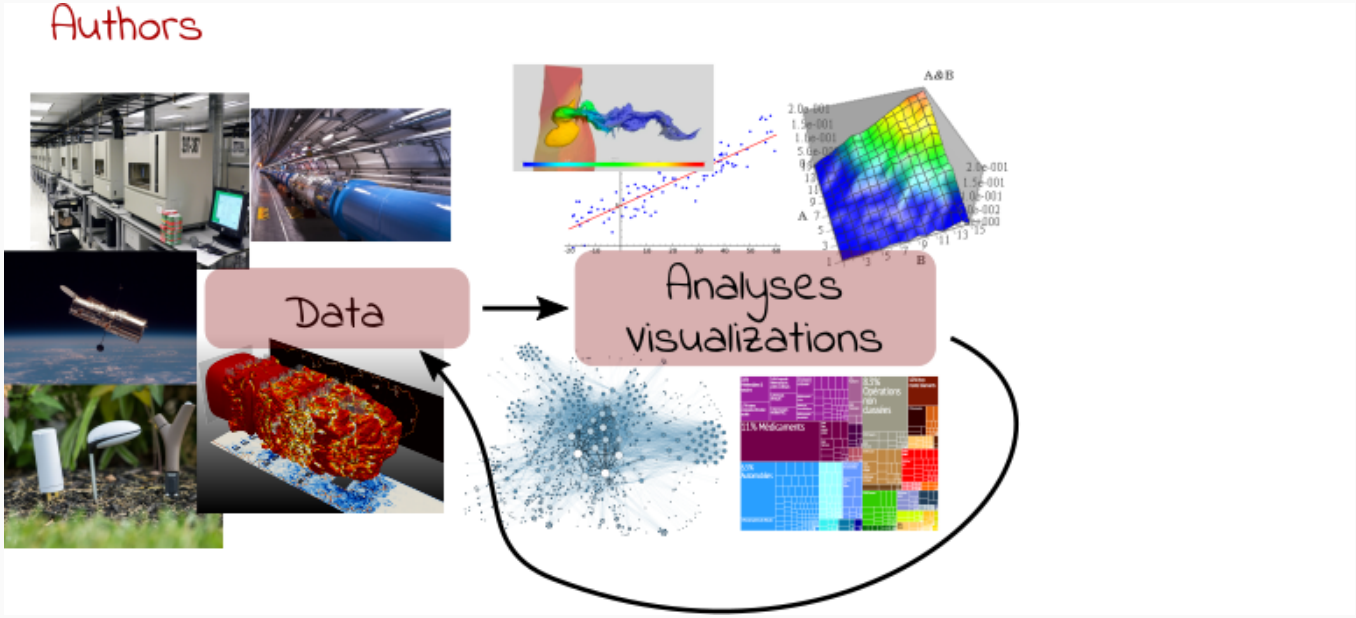
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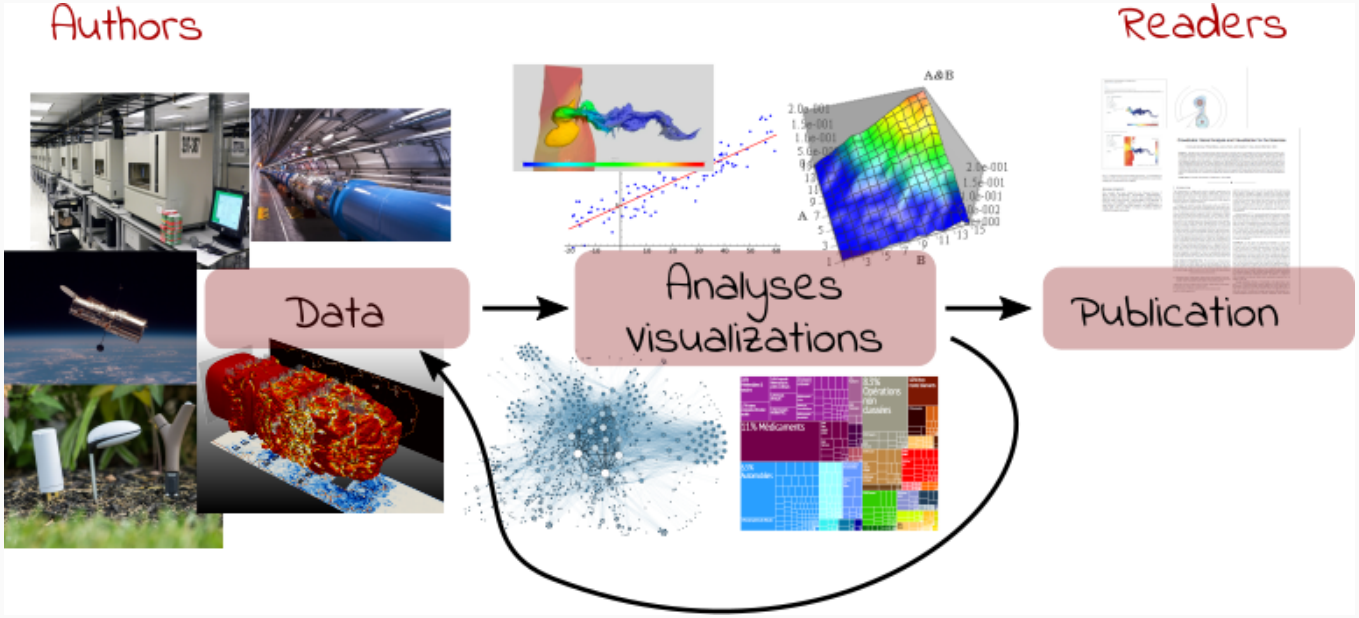
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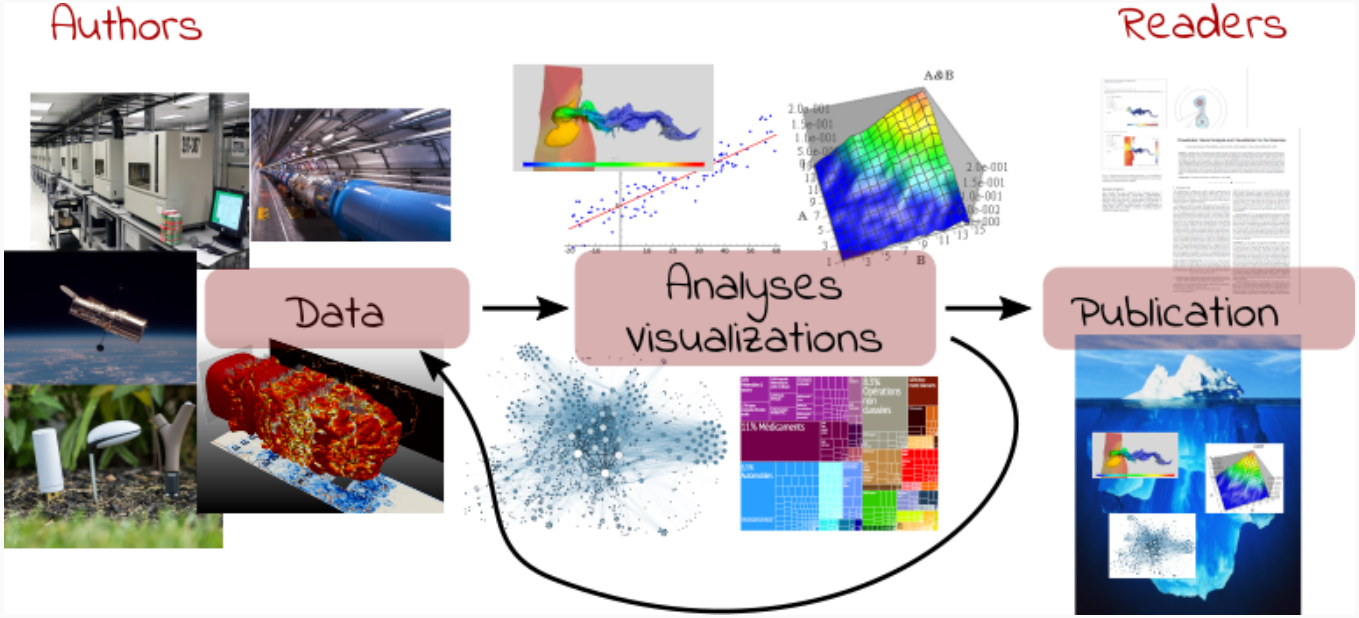
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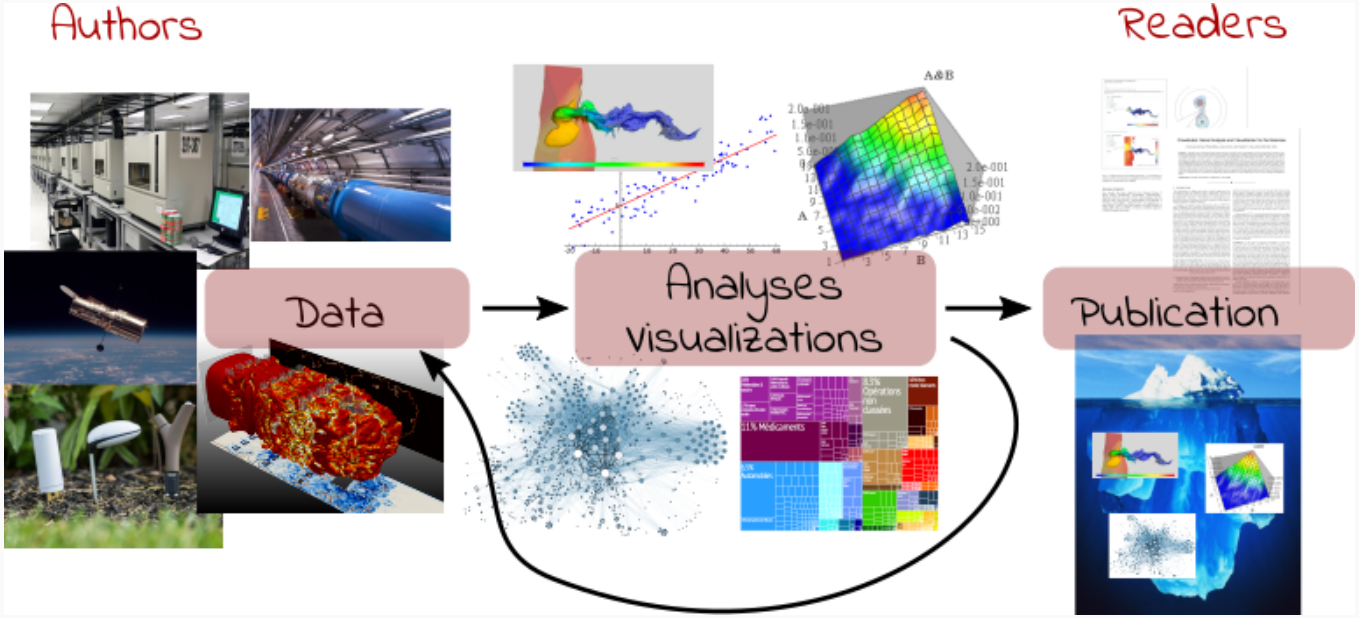
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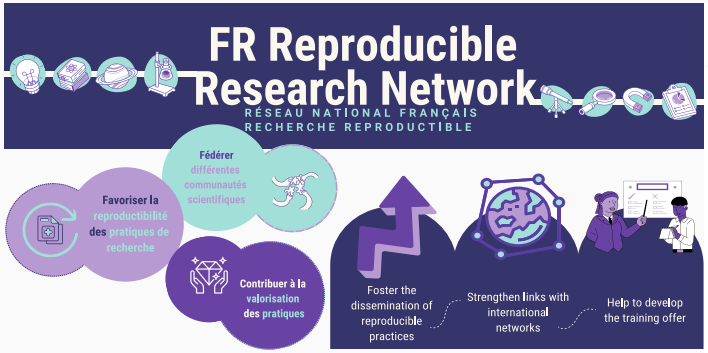


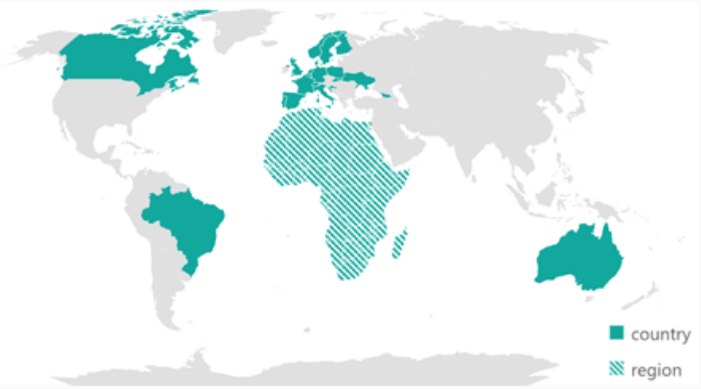
Reproducible Research = Bridging the Gap by working Transparently

Useful resources



A non-technical introduction to reproducibility issues

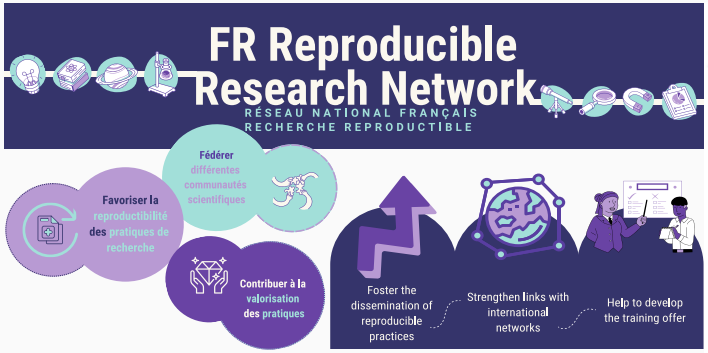


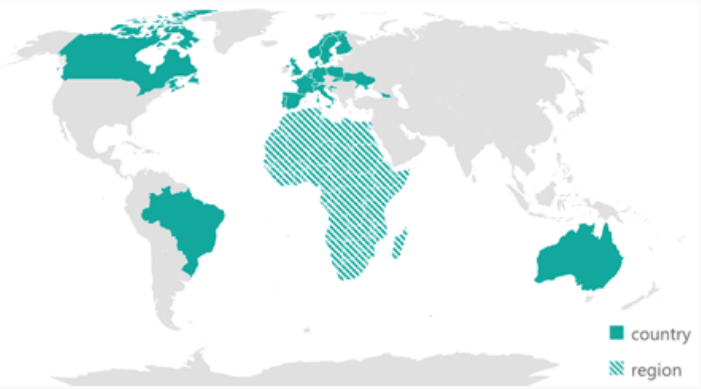


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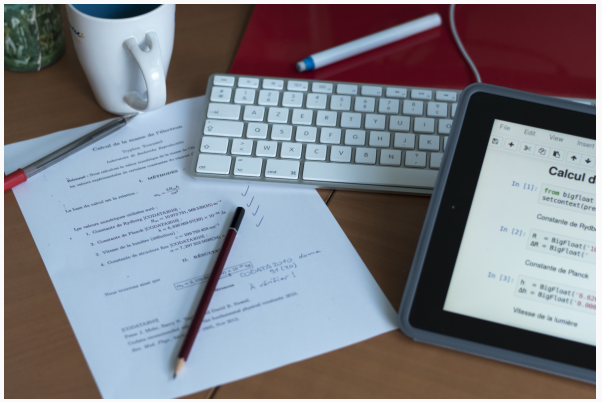
A non-technical introduction to reproducibility issues





MOOC Reproducible Research: Methodological principles for a transparent science, Learning Lab

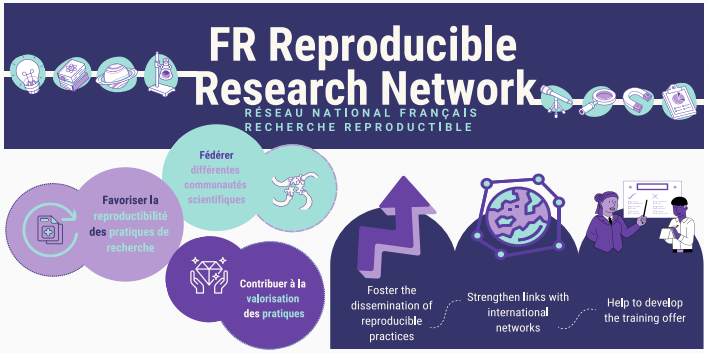
- Konrad Hinsen, Christophe Pouzat
- Laurence Farhi, Madeline Montigny, ...
- 2018 – ... (25,000+)

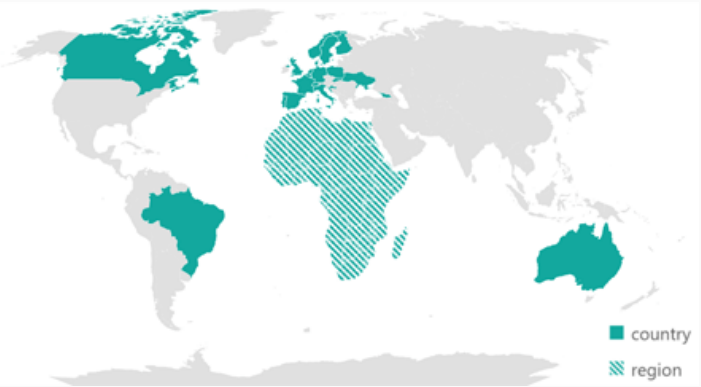


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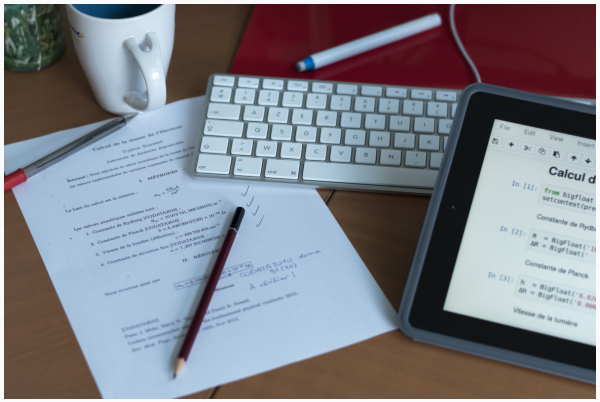
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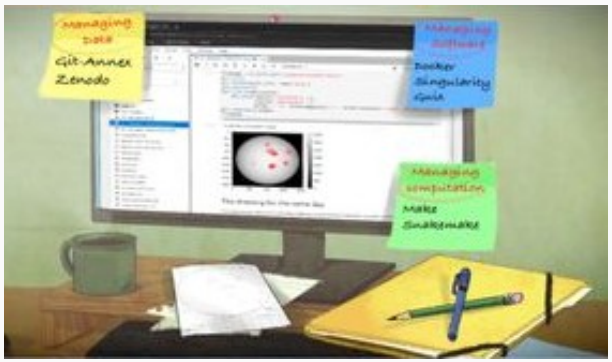




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- 2018 – … (25,000+)





MOOC RR2: Practices and tools for managing computations and data 2024 – … (3,000+)

- Managing data (`git annex`, Zenodo, SWH)
- Software environment control (`docker`, `singularity`, `guix`)
- Scientific workflow (`make`, `snakemake`)

Law evolution (in France)

- **2015**: EU puts forward as a priorities the *opening of science*, of *innovation* and *openness to the world* $\rightsquigarrow$ Societal science
- **2016**: Doctoral training on ethics and scientific integrity is mandatory (doctoral schools)
  - Related to the Corvol report (Bilan et propositions de mise en œuvre de la charte nationale d'intégrité scientifique)
- **2018**: National Plan for Open science
  - open access (HAL) and open data (FAIR)
  - sustainable and international movement



- **2021**: 2nd National Plan for Open science
  - 100% open access data by 2030
  - **open source/data/articles/ science is the new norm**
- **2022**: the ~~Hippocratic~~ PhD Oath

In the presence of my peers. Having completed my doctorate in [discipline], and having thus practiced, in my quest for knowledge, the exercise of a demanding scientific research, cultivating intellectual rigor, ethical reflexivity and respect for the principles of scientific integrity, I commit myself, for what will depend on me, in the continuation of my professional career, whatever the sector or the field of activity, to maintain an honest conduct in my relationship to knowledge, my methods and my results.

Three pillars for a Responsible Science

HCERES OFIS, CNRS COMETS, Inria COERLE, CERNA

1. Deontology
   - Le Pors 1983 rév. 2016: *"Le fonctionnaire exerce ses fonctions avec dignité, impartialité, intégrité et probité"*
     - Independence, equity, private activities, ... of civil servants
     - Conflicts of interest, connection w. industry, intel. property, harassment, ...
   - Law, deontology referent
2. Ethics
   - Deep questions raised by science progress on society; Moral values motivating our acts and their consequences
   - Regular discussions, connection with society [COMETS/COERLE/...]
   - Retrospective [law, approval] vs prospective view
3. Scientific Integrity
   - Rules governing research practices. Code of conduct for trustable research. Domain dependant
   - Universal, *soft law* professional code [Scientific integrity, whistleblowers referent]

Then came GenAI...

Timeline

- **Dec. 2022**: ChatGPT (OpenAI, Sam Altman and Elon Musk, Peter Thiel, AWS)
- **Feb. 2023**: LLaMA (Meta), open model
- **March 2023**: Claude (Anthropic, former OpenAI members, Dario and Daniela Amodei)
- **April 2023**: Qwen (Alibaba), based on LLaMA architecture
- **Dec. 2023**: Gemini (Google)
- **Feb. 2024**: Le Chat/Vibe (Mistral, Arthur Mensch)

These platforms/services are owned and operated by private entities acting on behalf of foreign powers (possible ingerence), and their use distorts academics capabilities and raises serious concerns about scientific integrity

Institutional Responses

- INRAE Guide to the Proper Use of GenAI-Based Assistants (Dec. 2025)
- CNRS "deploys" a sovereign AI assistant for research: Emmy (Dec. 2025)
- What about us ?! (Inria, CUMI-gra, June 2026)

The publication system collapse (1/2)

- **Dec 2025**: overwhelmed by AI-generated article submissions, arXiv enforces stricter moderation (no more survey).
- **May 2026**: Arxiv suspends researchers for one year if they submit erroneous AI-generated articles.
- **AAAI-26**: 29,000 submissions. After removing non fully compliant papers (missing PDFs, non-anonymized manuscripts, over-length papers, etc.), $\approx$ 23,000 papers under review, nearly × 2 compared to AAAI-25!
  AAAI-27: now asks for a payment for the second submission to slow down the LLM article injection rate.
- **ICLR'27**: submissions × 3 compared to ICLR'26 (× 2 compared to ICLR'25), from 490 Submissions to 62,000 in Ten Years
- **TMLR**: *The fraction of desk rejected papers at TMLR used to be about 6% in 2023 but is now at about 53%*
  (*Asking Authors About Their Own Papers*, by Nihar B. Shah, 09/26)

The publication system collapse (2/2)

- **NeurIPS 2026 AI-Assisted Reviewing Experiment**: *Except for this review experiment, NeurIPS does not authorize any other use of LLMs during the review process. Any such use constitutes a violation of integrity policies and may result in consequences for reviewers and their submitted papers, including desk rejection*
- **Nov. 2025**: A "bug" in the OpenReview platform led to a database leak in the **ICLR'26** conference submission and peer review system, which breached the double anonymous peer review process and made it possible for anyone to identify reviewers of specific submissions:
  - $\approx$ 21% of the 75,000 ICLR peer reviews were completely AI-generated
  - over 50% of the reviews showed some traces of AI use
  - nearly 200 of the submissions were found to be completely AI generated and 9% of the papers had over 50% of their text generated by AI

  Major AI conference flooded with peer reviews written fully by AI
- **Jan 2026**: **NeurIPS 2025**, 100 hallucinated citations in 51 accepted articles (out of 5,290, see gpt-0)
- **Sep 2026**: **ICLR 2027**, *We permit limited and responsible use of AI tools to assist human writing of peer reviews and meta-reviews* AI Policy

Publication growth is accelerating and it is not a good news!

from 6.8% (2016–2022) to 8.3% (2022–2025), with 2025 alone at +11.9%

Generative AI and Scientific Integrity (1/2)

Sep. 2026: Nicholas Polson (statistics, University of Chicago Booth) authored or co-authored 250+ papers on a diversity of topics and as many as 14 books released in a year (Washington Post) $\rightsquigarrow$ major retractation on the Social Science Research Network (The College Fix)

- Ivan Oransky (Retraction Watch): *it would be better to look at the system than Polson's specific case*
- Publish or perish, papermills, *tortured phrases* with SciGen, ... are not new (see the work of Cabanac and Labbé) and temporarily joining the dark side can be tempting when under pressure

Yet, Google AI coscientist, AI scientist, or AutoResearch fully align with the new US Kratsios Doctrine *Genesis : Science, a new golden age* (Jul. 2026)

- While Vannevar Bush (Roosevelt and Truman advisor) was laying in 1945 (*Science: The Endless Frontier*) the groundwork for a new universalism and a state-funded, open science, Kratsios seeks to reorganize science to serve American sovereignty and dominance. It is the exact opposite
- The *Gold Standard science* promoted by Trump in May 2025 and "rooted in the *reproducibility crisis*" is actually a smokescreen to get rid of "woke science" (social science, climate, etc.) and publicize pseudo-science

Scientific Misconduct

From https://en.wikipedia.org/wiki/Scientific_misconduct

Danish definition Intention or gross negligence leading to fabrication of the scientific message or a false credit or emphasis given to a scientist

Swedish definition *Intentional* distortion of the research process by fabrication of data, text, hypothesis, or methods from another researcher's manuscript form or publication; or distortion of the research process in other ways

US NSF definition Scientific misconduct revolves around the three following issues:

1. fabrication (completely made up results).
2. falsification (you obtain results and change them)
3. plagiarism (stealing someone else's sentence or idea without giving credit)

Data fabrication Examples

Dong-Pyou Han Assistant professor, Biomedical sciences, Iowa State University, 2013

Falsified blood results to make it appear as though a vaccine exhibited anti-HIV activity

- Han and his team received $\approx \$19$ million from NIH
- 1 retracted publication and resignation of university. Sentenced in 2015 to 57 months imprisonment for fabricating and falsifying data in HIV vaccine trials. $\$7.2$ million!

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Diederik Stapel Professor, Social Psychology, Univ. Tilburg, 2011

I failed as a scientist. I adapted research data and fabricated research. Not once, but several times, not for a short period, but over a longer period of time. [..] I am aware of the suffering and sorrow that I caused to my colleagues... I did not withstand the pressure to score, to publish, the pressure to get better in time. I wanted too much, too fast. In a system where there are few checks and balances, where people work alone, I took the wrong turn.

58 retracted publications

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Brian Wansink Professor, Psychological Nutrition, Cornell, 2016

I gave her a data set of a self-funded, failed study which had null results. I said "This cost us a lot of time and our own money to collect. There's got to be something here we can salvage because it's a cool (rich & unique) data set." I told her what the analyses should be. [..] Every day she came back with puzzling new results, and every day we would scratch our heads, ask "Why," and come up with another way to reanalyze the data with yet another set of plausible hypotheses

17 retracted publications

Data falsification

Table 2. Particle Sizes of IOCMCS/DNA Complexes Prepared in Water

charge ratio (N/P)	IOCMCS20 (nm)	IOCMCS 100 (nm)	IOCMCS200 (nm)
1	382.4 ± 12.1	392.1 ± 16.5	392.8 ± 15.4
10	394.1 ± 19.1	383.2 ± 15.4	200.2 ± 16.5
20	259.5 ± 20.3	306.1 ± 16.8	273.8 ± 17.5
30	349.3 ± 16.1	336.6 ± 21.5	214.7 ± 23.1
50	384.0 ± 14.2	306.5 ± 11.9	207.5 ± 21.9
80	363.4 ± 25.6	197.3 ± 15.6	206.7 ± 18.2
100	221.8 ± 14.8	205.3 ± 19.2	192.2 ± 16.8

Table 3. Particle Sizes of IOCMCS/DNA Complexes Prepared in PBS

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10	354.1 ± 15.1	393.2 ± 16.1	211.2 ± 15.2
20	289.5 ± 20.3	346.1 ± 17.5	293.8 ± 13.7
30	369.3 ± 17.1	386.6 ± 22.4	234.7 ± 21.3
50	394.0 ± 18.2	381.5 ± 12.8	227.5 ± 12.6
80	383.4 ± 15.6	297.3 ± 13.6	246.7 ± 19.4
100	241.8 ± 16.8	209.3 ± 21.2	195.2 ± 14.5

https://twitter.com/MicrobiomDigest/status/1416838153583927304

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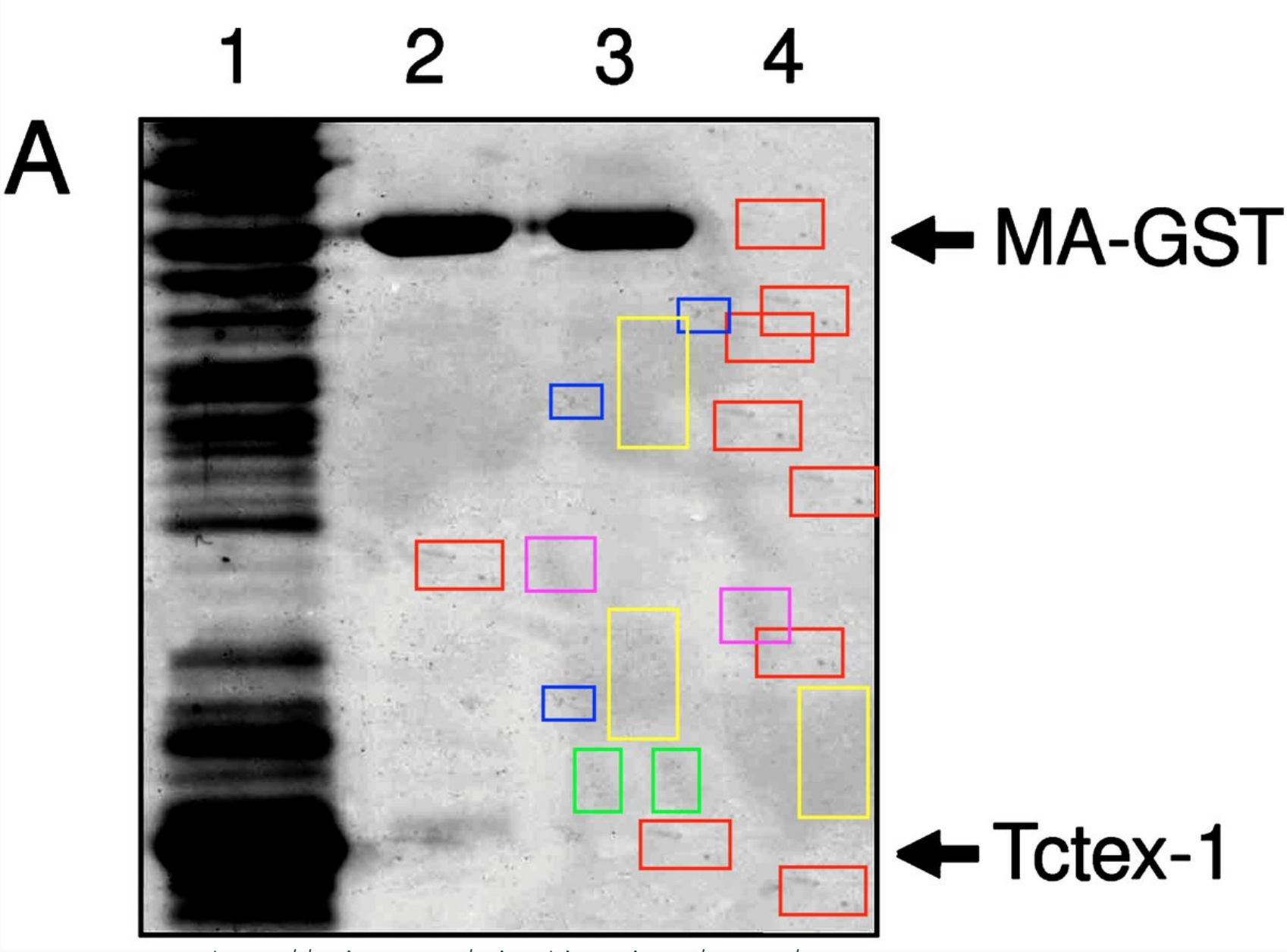
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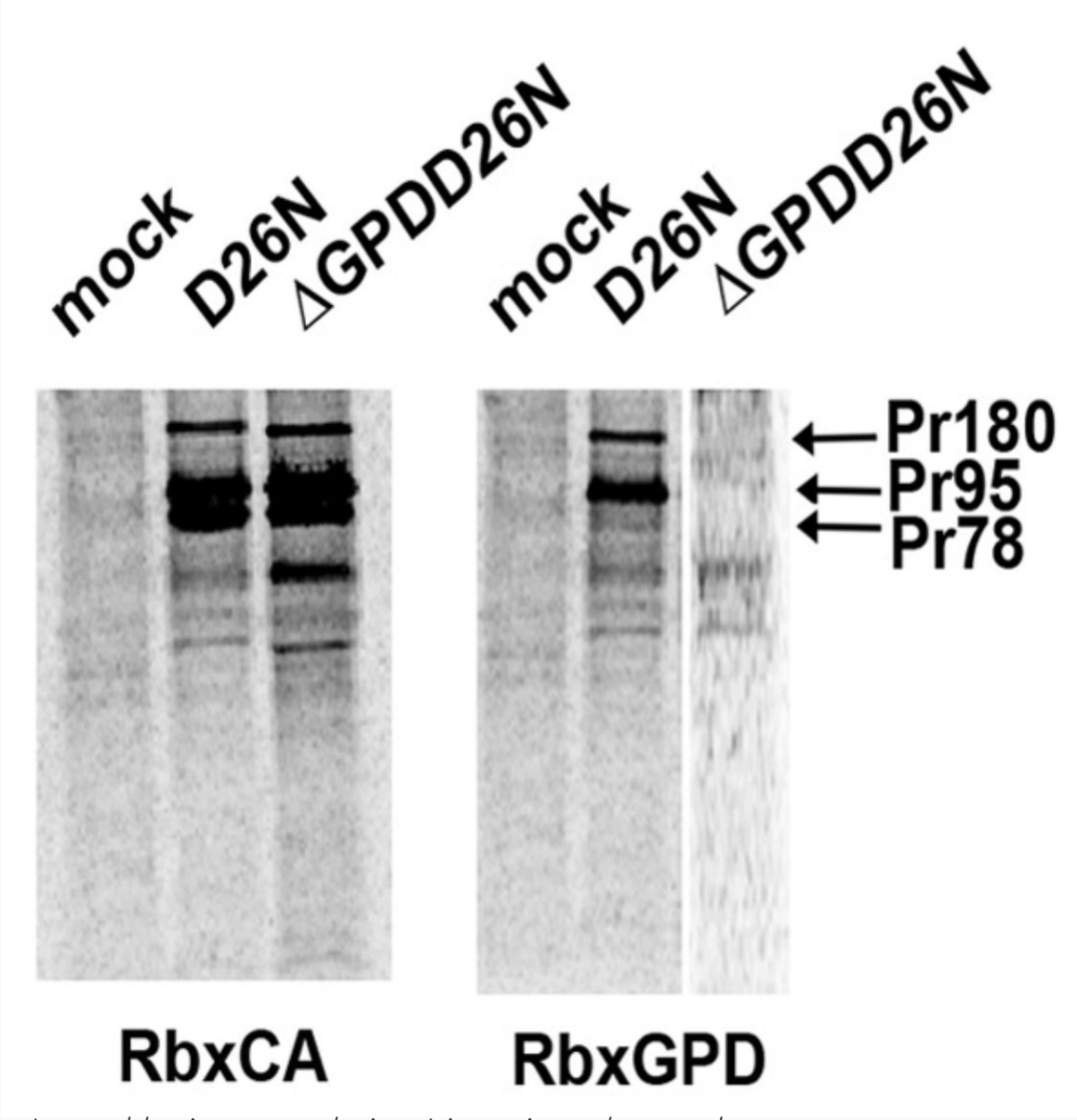
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Photo Manipulation



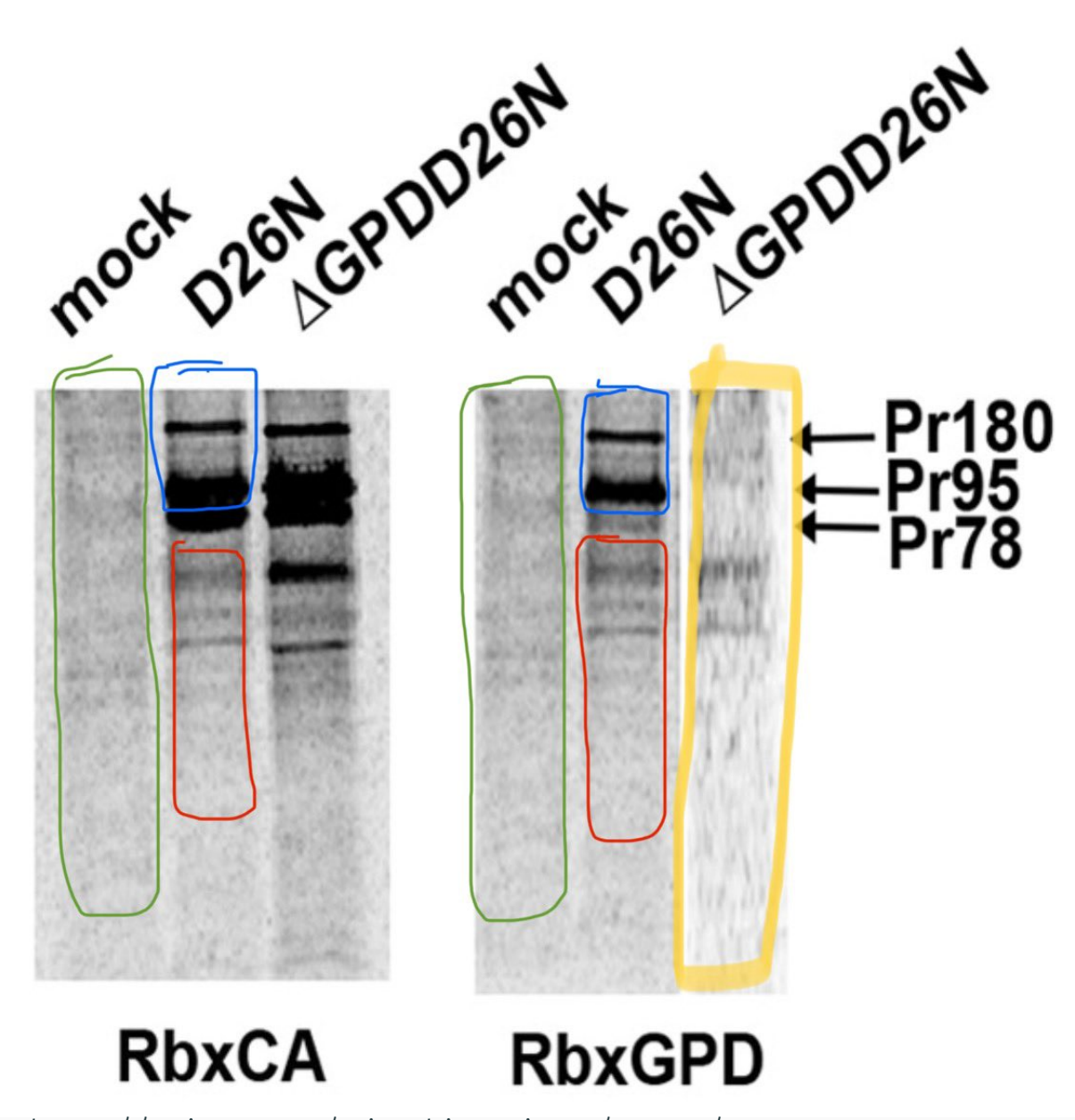
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Generative AI and Scientific Integrity (2/2)

Feb. 2024: Systèmes d'IA gén. : quelques points de vigilance (OFIS)

Responsibility Researchers remain responsible for the content generated by these systems and reproduced in their articles, including, among other things, the content's reliability and compliance with applicable regulations

Transparency Since June 2023, the European Code of Conduct for Scientific Integrity has recommended transparency. Concealing the use of AI in content creation or in the drafting of publications is now considered a breach of scientific integrity

Reliability GenAI systems make up information ("hallucinations"), which exposes to the diffusion of false information, data fabrication and falsification

Ownership GenAI content may be derived from training data protected by copyright. There is therefore a risk of plagiarism by users who appropriate this content.

- A GenAI system cannot be recognized as the author of an article or other research outputs (see "ChatGPT: Author of a Scientific Publication?)
- Details regarding the use of LLMs must be disclosed in the Methods section and/or the Acknowledgments

Maths under attacks (1/2)

- **Dec. 2025**: *Extremal descendant integrals on moduli spaces of curves: An inequality discovered and proved in collaboration with AI*, by Johannes Schmitt (ArXiv)

  > *The proof of the above result was found and formulated by the AI models GPT-5 and Gemini 3 Pro. Large parts of the paper were drafted by Claude Opus 4.5, and a part of the argument was formalized in Lean with the help of Claude Code and GPT-5.2. The paper aims for maximal transparency on the authorship of different sections and the employed AI tools (including prompts and conversation logs).*

- **April 2026**: Liam Price, a 23-year-old with no advanced mathematics training, was just "vibe maths-ing" with ChatGPT and solves Erdös problem #1196.

  in 80 minutes from a single prompt

  Btw, ChatGPT = 114.2 million users per day

Maths under attacks (2/2)

- **Sep. 2026**: Solution to the *Navier–Stokes existence and smoothness problem* by OpenAI (1 M$ Millennium Prize Problem)

  > *The group that produced the Navier–Stokes resolution involved on the order of 10,000 concurrent agents [..]. The agents arrived at their resolution on Saturday, September 5, about 88 hours after the first agents were launched. Lean formalization and verification took an additional 17 hours via GPT-6 Astra.*
  > *Across all attempted problems, the agents sent 4.9 million messages and used about 300 billion output tokens. In the process of resolving the Navier–Stokes problem, the agents sent 2.7 million messages and used approximately 130 billion output tokens (i.e., 10 M$ to 40 M$).*

  - Controversy: Tristan Buckmaster and Levent Alpöge claimed their own interaction with ChatGPT may have inspired OpenAI's prompts
  - This isn't science. It's a pissing contest between OpenAI and Anthropic. Maths isn't about solving problems (nor Science is about producing articles) but to understand things (David Madore)
- **October 2026**: OpenAI releases 722 mathematical papers
  - *"They've annihilated my field of research"* (Hugo Duminil-Copin)

This gives a double meaning to *Weapons of Math Destruction*

Ten Reasons why GenAI should not be used during PhD



Why you shouldn't use GenAI for your studies

Conscientious objection, AtEcoPol

1. Training and Cognitive Deficit **You're here to learn, not to produce**. AI may accelerate production but it undermines training: by letting AI think and produce for you, you're not exercising your own abilities
2. Satisfaction Succeeding on your own, without AI, brings pleasure and a sense of confidence in your own abilities, far more rewarding than verifying an AI generated content, which is demeaning and depressing
3. Reciprocity I will never use AI to create content, nor will I ever review articles using AI. Reading AI-generated content or realizing that it's been used it saddens me and undermines my trust
4. Paranoia and social withdrawal If anyone can solve hundreds of open problems using openly shared ideas, then quickly none will share anything and people will not exchange ideas anymore

Why I decided not to use GenAI at all

5. Working Conditions The growth of GenAI relies on the exploitation of hundreds of millions of clickworkers
6. Intellectual Property AI is based on the illegal appropriation of tens of millions of human artworks
7. Humanism GenAI is driven by Big Tech companies whose leaders are pushing an anti-humanist and anti-democratic agenda
8. Reliability An AI system always provides answers —including false or fabricated ones— and responds with problematic biases that replicate and amplify already existing ones (sexism, racism, etc.)
9. Data centers GenAI requires the construction of data centers, which have environmental impacts on nearby communities –often already disadvantaged– and lead to conflicts over the use of water and electricity
10. Global Warming GenAI contributes to the acceleration of global warming and destruction of ecosystems both directly and indirectly (through economic growth and deflecting attention)

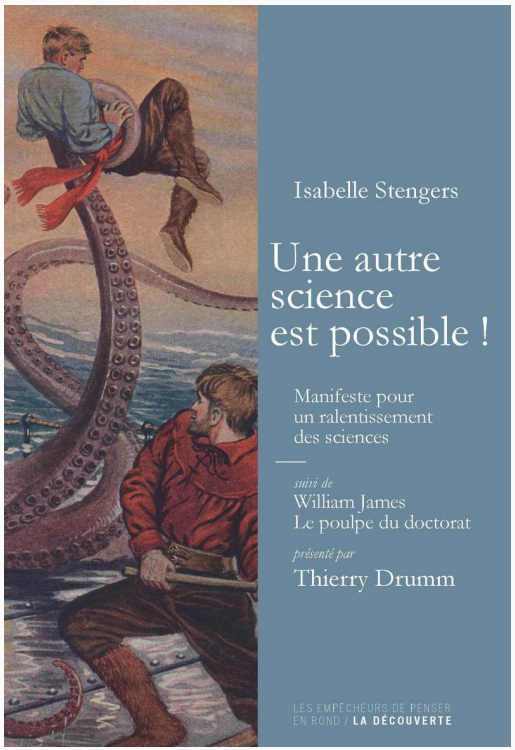
What do we do next ?

Complaining about this and conscientious objection ≠ reactionary movement. It is opening the debate about the ongoing Enclosure of the Scientific Commons

It is sad that Taylorism and proletarianization had to eventually affect even state intellectuals before they would care about the problem. Yet, Academia under attack - How do we speak up and organize?

In 2003: Let's Save Research (Sauvons la recherche)... of what? Liberal reforms? Herself?

I recommend reading *Une autre science est possible*, Isabelle Stengers, 2013.



That's all Folks!

Moving debate

1. Given your centers of interest and your practice, do you rather consider yourself as a mathematician or as a computer scientist?

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2. GenAI is mostly beneficial to my professional life

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2. GenAI is mostly beneficial to my professional life

3. The use of GenAI should be regulated